

CHAPTER 17

EXERCISE TECHNIQUE FOR ALTERNATIVE MODES AND NONTRADITIONAL IMPLEMENT TRAINING

Justin Keogh, PhD, and G. Gregory Haff, PhD

After completing this chapter, you will be able to

- understand the basic guidelines for performing resistance exercise with alternative modes and nontraditional implements;
- describe the benefits and limitations of bodyweight training activities;
- identify the benefits and limitations associated with core training;
- identify the appropriate technique and key technical flaws associated with the alternative mode exercises;
- appropriately determine how to apply resistance bands and chains to traditional ground-based free weight exercise; and
- determine the appropriate use of alternative methods and nontraditional implement exercises.

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The use of alternative modes and nontraditional implement exercises has become increasingly popular in the strength and conditioning profession. Whenever implementing these types of training methods into training programs, one should consider some basic and specific guidelines to ensure that these methods are used safely.

General Guidelines

With the use of **alternative modes** and **nontraditional implement** exercises, the general guidelines are not very different from those used with traditional resistance training methods. A stable body position that allows the athlete to achieve and maintain safe and proper body alignment during the performance of the exercise is needed to appropriately stress the skeletal muscle system. Freestanding ground-based exercises typically use a position in which the feet are placed slightly wider than shoulder width. With the use of instability devices, the body position may need to be modified to ensure that stability is achieved. The grip used with alternative modes and nontraditional implement exercises is typically one of the traditional grips presented in [chapter 16](#). The choice of grip is based on the demands of the particular exercise. Additionally, with many nontraditional implements, the grip can be a limiting factor in the performance of the exercise.

As with more traditional exercises, the breathing pattern often recommended with these alternative methods is for the athlete to exhale through the **sticking point** (concentric portion) and to inhale during the less stressful portion (eccentric portion) of the movement. For example, athletes doing the dumbbell chest press on a stability ball would inhale as they lower the dumbbells to the chest and exhale as they push them away from the chest. With structural exercises (those that load the **axial skeleton**), breath holding may be warranted. However, when one is lifting loads greater than 80% of maximal voluntary contraction or lifting lighter loads to failure, the **Valsalva maneuver** (forced expiration against a closed glottis) may be unavoidable (19, 47). The Valsalva maneuver allows for an increase in the intra-abdominal pressure that augments the stability of the spine, which may be beneficial for performing nontraditional exercises. For example, with the log clean, the athlete may perform the Valsalva maneuver during the pull and catch portion of the exercise. The athlete would then exhale after assuming an erect

position. [Chapter 16](#) presents more information on the Valsalva maneuver.

Bodyweight Training Methods

Bodyweight training methods are among the most basic methods for performing resistance training. Specifically, with these types of exercises, the body weight of the individual is used to provide resistance (56). Activities such as push-ups, pull-ups, chin-ups, sit-ups, and squat thrusts are typically mentioned in the context of bodyweight training. However, activities such as calisthenics, gymnastics, and yoga could all be classified as bodyweight training methodologies (56). As noted by Behm and colleagues (14), gymnastics was classically a part of the physical education system; this type of training strongly promotes the development of the core musculature, which, when strengthened, appears to reduce injury potential. Bodyweight training offers a low-cost training method that allows for the development of relative strength levels.

One of the issues related to bodyweight resistance training is the fact that the resistance load is limited to the individual's body weight. As such, bodyweight training tends not to significantly affect absolute strength levels (56). To increase the intensity of bodyweight exercises, one can do several things, including increasing the number of repetitions or changing the movement pattern. While increasing the number of repetitions will change the workload, it will shift the targeted outcome from strength toward strength endurance, which shifts the training target away from the development of strength. Simple modifications to bodyweight exercises may obviate some of these limitations. Changing the movement pattern (e.g., elevating the legs while doing a push-up) will increase the resistance encountered.

General Guidelines for Alternative Modes and Nontraditional Implement Exercises

- Ensure proper body alignment via selection of a stable body position.

- If the exercise is a freestanding ground-based exercise, place the feet slightly wider than shoulder width and keep them flat on the ground.
- Use the appropriate grip for the exercise based on the type of exercise (see [chapter 16](#) for more detail).
- Exhale during the concentric portion of the exercise and inhale during the eccentric phase.
- With heavy loads (80% of maximal voluntary contraction or greater) or with lighter loads performed to failure, the Valsalva maneuver may be a useful technique for maintaining spinal stability.

Benefits of Bodyweight Training

The following are some of the benefits of bodyweight training (56):

- Is specific to each individual's anthropometrics
- Often includes closed-chain activities that engage multiple joints
- Strengthens several muscle groups simultaneously
- Develops relative strength
- Improves body control
- Is a versatile form of training that can be done anywhere
- Is a low-cost training alternative

Suspension training is an alternative method for performing bodyweight resistance training that allows for the modification of the resistance applied during numerous upper and lower body exercise options (90). Performing this type of training uses devices, referred to as *suspension trainers* (e.g., TRX®), that are typically constructed with sturdy and adjustable straps that have handles or foot straps attached to one end. These devices are generally suspended from a singular or multiple centralized attachment points above the user's head (e.g., wall, squat rack, doorjamb, ceiling) allowing the straps to hang perpendicular to the floor (31, 113). Athletes can hang from the device by holding the strap handles or placing their feet in the foot straps to manipulate the placement of their center of mass, which allows for the intensity of the exercise to be manipulated (139).

The degree of resistance encountered with this type of training is dictated by altering the angle of the suspension trainer's straps, effectively changing the gravitational pull as a percentage of the individual's body weight (90, 112). Generally, as the athlete's center of mass moves farther away from the anchor point, the lower the amount of resistance to which the athlete is exposed (90). For example, Melrose and Dawes (90) report that when the inverted row is performed with the straps at a 75-degree angle from the ground (i.e., center of mass closer to the attachment point) the intensity of the exercise will correspond to $79.4\% \pm 2.1\%$ of body weight. If, however, the angle is changed to 30 degrees from the ground (i.e., center of mass is farther away from the attachment point), the resistance encountered is reduced to $37.4\% \pm 21.5\%$ of body weight. It is clear based on this study that suspension training is limited in its ability to provide resistances greater than body weight; however, it is important to note that resistance can be added by wearing a weighted vest during suspension training, thus increasing the potential training stimulus (30, 36).

While there are limitations to the amount of load that can be applied when performing suspension training, this type of training has several benefits (2, 30, 36, 90), including the following:

- *Versatility.* Can be used with a wide range of exercises that target a variety of muscle groups
- *Portability.* Is easily transported when traveling and can be used indoors or outdoors
- *Space-efficient.* Requires minimal storage space
- *Cost-effective.* Relatively inexpensive compared to other resistance training methods
- *Scalability.* Can be adjusted to increase or decrease difficulty through changing body position or angle
- *Balance and coordination.* Challenges proprioception and body awareness
- *Muscle activation.* Changes in body position can alter muscle activation patterns

- *Adaptability*. Can be used with a variety of populations

Core Stability and Balance Training Methods

There is increasing interest in training the **core** in an attempt to improve overall health, rehabilitate from injuries, and enhance athletic performance (14). To target core stability and balance, interventions ranging from traditional **ground-based free weight exercises** to training on unstable devices have been advocated in the applied literature.

Anatomical Focus

The term *core* is commonly used in the popular media and in some training journals (13) to refer to the trunk or, more specifically, the lumbopelvic region of the body (151). However, interpretation of the scientific literature appears to make an accurate or consistent definition of the core tenuous at best (13, 153). Typically, the **anatomical core** is defined as the axial skeleton and all the soft tissues with proximal attachments that originate on the axial skeleton (13, 14). It is important to note that the appendicular skeleton includes the pelvic and shoulder girdles, while the soft tissues include the articular cartilage and fibrocartilage, ligaments, tendons, muscles, and fascia (13). Ultimately, the soft tissues act to generate forces (concentric muscle actions) and resist motion (eccentric and isometric muscle actions).

The muscles that are typically associated with the core allow for the transference of torques and angular momentum during the performance of integrated kinetic chain activities, such as kicking or throwing (111, 151). In fact, Willardson (151) suggests that increasing an athlete's core stability will result in a better foundation for force production in the upper and lower extremities. Additionally, Rodriguez-Perea and colleagues (111) report that increasing core stability can improve balance outcomes, throwing or hitting velocity,

throwing or hitting distances, vertical jump performance, and horizontal jump performance.

Isolation Exercises

Isolation exercises typically consist of dynamic or isometric muscle actions designed to isolate specific core musculature without the contribution of the lower and upper extremities (14). For example, common exercises performed to isolate the core are the prone plank (118) and side plank (140). Evidence shows that these types of isolated exercises can increase **muscle activation**, which has been suggested to result in improvements in spinal stability and a reduction of injuries (94). While evidence suggests that these types of activities result in improvements in performance in untrained individuals and those recovering from an injury, support is limited for the idea that these types of training activities translate to improvements in sport performance (81, 109, 151). While Lou and colleagues (81) have reported a positive transfer between the effects of core training on basketball-related performance, several other authors have reported that isolated core training is not very effective at improving sport performance (14, 109, 151). Additionally, according to Behm and colleagues (14) and Willardson (151), there is strong evidence that ground-based free weight exercises (e.g., squat, deadlift, push press, snatch, and exercises that involve trunk rotation) offer a greater benefit to actual sport performance compared to isolated core training. Ground-based free weight activities appear to offer activation of the core musculature that is similar to, or in most cases greater than, what is seen with traditional isolation exercises designed to engage the core (50, 102). Isolation exercises may have the greatest benefit for injured athletes who are going through the rehabilitation process and are not able to adequately load traditional ground-based free weight exercises (151).

Ground-based free weight activities appear to offer similar or, in most cases, greater activation of the core musculature when compared to traditional isolation exercises designed to engage the core.

Machines Versus Free Weight Exercises

When free weight training methods are compared to machine-based training methods, each has advantages and disadvantages (14, 49, 58, 107, 128). With regard to **machine-based training**, the stability provided by the machine may result in a better ability to target specific muscle groups, which may be useful when targeting hypertrophy (120), but in the context of sport performance, muscle rarely if ever functions in such an isolated fashion (14). With regard to the stabilizer muscles, it is generally accepted that activation of these muscles is greater during free weight training when compared with machine-based training (48). In partial support of this belief, Anderson and Behm (6) state that the activity of the back stabilizers was 30% lower during a Smith machine squat when compared to a free weight squat. Additionally, there is scientific support for the contention that machine-based strength gains exert a negligible or a possible detrimental effect on the muscle activation pattern in athletic movements (6, 20, 97). However, if instability is increased via performance of ground-based free weight training on unstable surfaces or devices, greater decrements in force production, the rate of force development, and power outputs have been noted (33, 84). Therefore, based on these data, it appears that free weight ground-based exercises offer the ideal combination of specificity and instability, especially when one is focusing on strength and power development. Ultimately, sufficient instability for the development of sport-specific adaptations can be stimulated via the use of traditional ground-based free weight strength training, and there appears to be no need to add increased instability to these types of exercises (14).

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Instability Devices

Instability-based exercises are typically considered those that are performed on unstable surfaces or devices commonly found in strength and conditioning facilities. The increased popularity of training on instability devices seems to have stemmed from their use by physiotherapists in the rehabilitation process. These devices are used to promote postural disequilibrium or imbalance requiring a greater stabilizing function of the core musculature (14, 151). If perturbations are applied during use of these devices, a balance challenge can occur that requires activation of the core musculature in order to induce postural adjustments to remain upright (27).

Numerous instability devices are available to the strength and conditioning professional; the most common are the Swiss, physio, and Pezzi gymnastics balls (14). Other instability options include hemispherical physioballs with one inflated dome side and a flat rubber side, inflatable disks, wobble boards, balance boards, foam tubes, and various foam platforms. Natural surfaces such as sand can also create scenarios that introduce instability into an athlete's movement pattern, creating an increased balance challenge that results in increased core muscle activation. Many strength and conditioning professionals believe that performing instability exercises trains the target agonist muscle group while simultaneously increasing core muscle activation (14). While some evidence suggests that core muscle activation may be increased, it appears that this increase occurs in conjunction with reduced force generation by the agonist muscle (13, 33). In fact, as the degree of instability of the exercise increases, there is a concomitant reduction in force-generating capacity (76). During training with instability exercises, the overall agonist force-generating capacity (12) and

overall power output (33) may be less than 70% of what can be achieved when the exercise is performed under stable conditions. Additionally, there can be a significant reduction in the rate of force development that is achieved during the exercise (84). Training with a reduced force production, power output, and rate of force development may not be the most advantageous method for preparing athletes, because these factors have a large impact on many aspects central to successful sport performance.

Overall, performance of resistance exercises on unstable devices by athletes results in muted performance improvements when compared to training on stable devices (29, 126, 161). The lack of performance benefit noted in the literature might be expected based on the lack of dynamic correspondence between exercises performed on unstable surfaces and sporting activities (45, 129). Additionally, the ineffectiveness of training on unstable surfaces may also be related to the principle of diminishing returns, because it is very likely that trained athletes require a much greater adaptive stimulus in terms of force production, movement velocity, and rate of force development to realize performance gains than can be provided by instability exercise devices (14, 79). Therefore, the performance of static balance activities on instability devices may be considered an introductory training step to improve balance and core stability before the implementation of dynamic or explosive ground-based free weight exercises, such as the Olympic lifts and their derivatives, that are performed on stable surfaces (14, 161).

Ground-based free weight exercises (e.g., squats, deadlifts, Olympic lifts and their derivatives) involve a degree of instability that allows for simultaneous development of all links of the kinetic chain, offering a much better training stimulus for the development of core stability and the enhancement of athletic performance than do instability device-based exercises.

When used in the rehabilitation setting, unstable devices have been shown to reduce low back pain and improve the efficiency of the soft tissues that stabilize the knee and ankle joints (13, 14). Since several muscles related to the knee joint originate in the lumbopelvic region, the core of the body could be considered an important contributor to the prevention of **anterior cruciate ligament (ACL)** injuries (99). In fact, several studies provide evidence that using instability devices may reduce the likelihood of ACL injuries, especially after rehabilitation from an ACL injury (100). For example, Fitzgerald and colleagues (40) reported that activities that challenge the rehabilitating athlete using perturbations stimulated by tilt boards, roller boards, and other balance devices make a successful return to competitive sport activities five times more likely. Historically, it is often suggested that the addition of balance training to traditional training methods results in a reduction in ACL injuries (24). However, in a systematic review, Grimm and colleagues (46) contest this contention and suggest that these types of interventions do not reduce ACL injury risk. In contrast, ground-based free weight activities seem to have a more balanced effect, leading to improvements in performance while increasing core strength and balance abilities. Ultimately, the use of instability-based exercises to train the core seems to be an effective method for returning the injured athlete to competitive-based training but provides a suboptimal training stimulus and provides little benefit for healthy uninjured athletes.

Variable-Resistance Training Methods

Resistance training practices include three methods for applying overload to the body: **constant external**, **accommodating**, and **variable resistance** (41, 88). The most common method for applying resistance during training is to use a constant external load, which is best represented by traditional resistance training methods (e.g., free weights). In this scenario, the external load remains constant throughout the range of motion, better represents real-life activities, and allows for a more realistic skeletal muscle coordination and movement pattern (54, 88, 106). On the other hand, accommodating resistance (sometimes referred to as *semi-isokinetic resistance applications*) generally allows for the speed of movement or isokinetic resistance to be controlled throughout a range of motion (88, 128). Stone and colleagues (128) suggest that these types of devices have poor external validity. Additionally, these devices are unlikely to provide an adequate training stimulus when compared to more traditional methods such as those involving constant-loaded free weight movements, particularly when those movements are performed with multijoint movement patterns.

With traditional resistance training exercise, the external load remains constant, but the forces exerted by the muscle vary in accordance with the mechanical advantages associated with the joints involved in the movement (17, 41, 130). In order to combat the changing mechanical advantages and inertial properties associated with constant-loaded resistance, a concerted effort has been made to develop innovative training devices that allow the applied resistance to be varied in conjunction with changes in joint angle (41, 145). These variable-resistance methods attempt to alter the resistance, so the muscle maximizes force application throughout the full range of motion (130, 145). For example, in the back squat the greatest muscle force production occurs at the midpoint of the movement, while the smallest forces are produced at the bottom. Thus variable-resistance methods would be used to reduce

resistance at the bottom of the squatting motion and increase resistance as the athlete ascends from the bottom position. Another consideration is that during the concentric effort of a movement, a large portion of time is spent decelerating (41). Overall, it has been suggested that variable-resistance methods may be able to match the changes in joint leverage (164), overcome the mechanical disadvantages associated with specific joint angles (35, 123, 124, 146), and provide for compensatory acceleration (123, 124).

With the use of a variable-loaded resistance model, the methodology most commonly seen in modern strength and conditioning facilities is the application of chains or rubber bands (41, 89, 133). The combination of chains or bands with traditional free weight resistance training methods has been shown to alter the loading profile typically seen during these constant-loaded activities (41, 66). Specifically, the chains or bands allow for the resistance to be varied across the range of motion achieved during the activity.

With use of a variable-loaded resistance model, the most common methodology used in modern strength and conditioning facilities is the application of chains or rubber bands.

Chain-Supplemented Exercises

One increasingly popular method of applying variable resistance is the addition of **chains** to traditional resistance training activities such as the bench press or back squat (8, 17, 59, 88, 145). This method of force application is most popular among powerlifters (134) but has become increasingly popular among strength and conditioning professionals working with a variety of sports (149). Despite the increasing popularity and the belief that these methods provide a training advantage, these beliefs are largely unsubstantiated in the scientific literature (17, 18, 59, 88, 130). Some studies, however, demonstrate that the application of chains to traditional resistance training methods such as the bench press (7,

10) can be advantageous. Careful inspection of these studies reveals that the means by which the chains are applied to the free weight exercise may influence their effectiveness. Specifically, these studies used a method in which the chain was suspended from the bar without touching the floor until the athlete had reached the lowest position in the squat or until the bar had reached chest height in the bench press (7, 10). While some research seems to support this methodology, much more research is needed to explore the various methods of applying chains to traditional resistance training methods.

Determining Resistance With Chains

The resistance provided by chains is largely dictated by the structure, density, length, and diameter of the chain and must be quantified before the chain is used in a resistance training setting. Additionally, the number of links in a chain will affect the amount of resistance provided by the chain (17, 89). To quantify the loading provided by chains, Berning and colleagues (17) developed a practical chart that related chain link diameter and length to the resistance load provided by the chain. This chart was later modified by McMaster and colleagues (88) to show the relationship between the chain mass, length, and diameter ([table 17.1](#)).

TABLE 17.1 Mass, Length, and Diameter of Chains

Chain diameters	Chain lengths				
	10 CM (4 IN.)	50 CM (20 IN.)	100 CM (39 IN.)	150 CM (59 IN.)	200 CM (79 IN.)
6.4 mm (1/4 in.)	0.3 kg (0.7 lb)	1.3 kg (2.9 lb)	2.5 kg (5.5 lb)	3.8 kg (8.4 lb)	5.0 kg (11.0 lb)
9.5 mm (3/8 in.)	0.4 kg (0.9 lb)	1.9 kg (4.2 lb)	3.7 kg (8.2 lb)	5.6 kg (12.3 lb)	7.4 kg (16.3 lb)

Adapted by permission from D.T. McMaster, J. Cronin, and M. McGuigan, "Forms of Variable Resistance Training," *Strength and Conditioning Journal* 31, no. 1 (2009): 60

12.7 mm (1/2 in.)	0.7 kg (1.5 lb)	3.7 kg (8.2 lb)	7.4 kg (16.3 lb)	11.1 kg (24.5 lb)	14.8 kg (32.6 lb)
19.1 mm (3/4 in.)	1.4 kg (3.1 lb)	7.0 kg (15.4 lb)	14.0 kg (30.9 lb)	21.0 kg (46.3 lb)	28.0 kg (61.7 lb)
22.2 mm (7/8 in.)	2.2 kg (4.9 lb)	10.8 kg (23.8 lb)	21.6 kg (47.6 lb)	32.4 kg (71.4 lb)	43.2 kg (95.2 lb)
25.4 mm (1 in.)	2.8 kg (6.2 lb)	14.0 kg (30.9 lb)	28.0 kg (61.7 lb)	42.0 kg (92.6 lb)	56.0 kg (123.5 lb)

Adapted by permission from D.T. McMaster, J. Cronin, and M. McGuigan, "Forms of Variable Resistance Training," *Strength and Conditioning Journal* 31, no. 1 (2009): 60

Determining the Load to Use With Chains

To determine the load used with chains, the absolute chain resistance at the top and at the bottom portion of the movement are summed and then averaged. For example, if athletes wanted to train at a 5-repetition maximum (5RM) load in the bench press, they would first determine the 5RM load without the chains. Then, if their 5RM is 120 kilograms (265 lb), they would subtract the average chain resistance from this load. If at the bottom position the load is 0 kilograms and at the top the chain load is 11.1 kilograms (24.5 lb), the average is 5.55 kilograms (12.2 lb). Thus, the athlete would add 114 to 115 kilograms (251.3-253.5 lb) to the barbell to achieve the appropriate loading.

As a means of deciding on the barbell resistance to use in conjunction with chains, the absolute load is determined for the top and the bottom portion of the movement (8). The average of these two loads is then calculated and used to modify the barbell load in order to allow the athlete to train in the prescribed range.

As a general rule, Baker (8) recommends that the use of chains be reserved for experienced intermediate- and elite-level athletes who have stable exercise technique, because the addition of chains provides a loading challenge that can affect the athlete's technique.

Applying Chains to Free Weight Exercises

Generally, the application of chains to traditional resistance training methods allows for a linear increase in the applied resistance (88).

Ways to apply chains include letting them touch the floor from the fully extended position during the movement (17) or hanging them from lighter chains ([figure 17.1](#)), which allows them to touch the floor only on reaching the lowest portion ([figure 17.2](#)) of the movement pattern (i.e., bottom of the squat or at chest level during the bench press) (8, 10). Baker (8) suggests that the second method may affect the velocity of movement in three distinct ways. First, the total barbell–chain complex comes into play only at the top of the movement (i.e., extend portion) when the chain links have been lifted off the floor. At the bottom of the movement, the links are in full contact with the floor, providing a reduction in load and allowing the athlete to accelerate the barbell at a faster rate. Second, it is possible that a within-repetition postactivation potentiation effect may occur in response to a greater neural activation. Specifically, when the chains pile on the floor and the mass of the barbell decreases, a greater neuromuscular activation may occur, allowing for an enhancement in movement velocity. Finally, it is possible that the decreasing resistance at the bottom portion of the movement may cause a more rapid stretch–shortening cycle. Baker (8) suggests that this happens in response to the eccentric unloading that occurs when the chain links pile on the floor at the bottom of the movement and a quicker amortization phase occurs when the athlete shifts from eccentric to concentric muscle action.



FIGURE 17.1 Bench press with added chain resistance. The links of the chains are draped over the lighter supporting chains, such that the chain does not contact the floor until the barbell reaches chest height. As the athlete lowers the barbell to the chest, the resistance progressively decreases as the chains furl onto the floor. Conversely, as the athlete moves the barbell concentrically away from the chest, the resistance progressively increases as the chains unfurl from the floor.



FIGURE 17.2 When the barbell is at chest height during the bench press, the chains are furling on the floor, with the result that their weight does not act on the barbell.

Resistance Band Exercises

The use of **resistance bands** to augment traditional barbell resistances has become increasingly popular among strength and conditioning professionals (8, 39, 77, 138, 145). There is some research support for the use of resistance bands combined with

traditional resistance training exercises (5, 104, 127, 146). For example, Wallace and colleagues (146) suggest that using bands to substitute 35% of the total load during a back squat can acutely increase peak power by approximately 13%. Additionally, Baker and Newton (9) suggest that the use of bands may result in a postactivation potentiation effect within each repetition. Support for this contention comes from the work of Stevenson and colleagues (127), who demonstrated that the use of bands to account for 20% of the training load resulted in an acute increase in the concentric rate of force development when compared to a constant-loaded condition. While these data seem to suggest that the addition of bands to traditional resistance training methods may offer some benefit, especially to power or rate of force development (66), some research contests these findings (35, 115). For example, Ebben and Jensen (35) report no differences in integrated electromyography and mean ground reaction forces when resistance bands are used to apply 10% of the total loading during squats.

Currently, minimal research on the use of resistance bands, especially longitudinal studies, has been conducted to determine whether the acute effects noted by some authors translate into superior strength and power gains over time. However, there is some evidence that the use of resistance bands during an off-season strength training program performed by Division 1AA football players induces a greater increase in upper body strength when compared to performing traditional bench press training (44). While these data are promising and provide some support for using bands as part of a performance-based resistance training program, much more research is needed to understand how these methods can be integrated into a periodized training program.

Determining Resistance With Resistance Bands

In working with resistance bands, it is important to understand that their composition varies depending on the type of thermoplastic or elastomer used to produce them (88). The composition of resistance

bands is important because it can exert an impact on their overall stiffness, density, yield, and tensile strength (82, 88, 89, 138). Ultimately, the tension (resistance) generated by a resistance band is determined by the overall stiffness of the band and the extent to which the band is stretched (deformation) (89, 165). Specifically, based on Hooke's law, the tension generated by a band is equal to its stiffness (k) multiplied by the deformation (d):

$$\text{Tension} = \text{Stiffness } (k) \times \text{Deformation } (d)$$

As the band is stretched (i.e., increased deformation), the amount of tension placed on the band linearly increases. However, several studies suggest that resistance bands exhibit both curvilinear and linear deformation regions (5, 89, 105).

When using bands in conjunction with traditional free weight training exercises such as the bench press or back squat, the strength and conditioning professional must be aware that there can be a 3.2% to 5.2% difference between two supposedly equal bands that could result in an 8% to 19% difference in mean tension between the bands (89). A basic length–tension relationship can be seen in [table 17.2](#), and tension can be predicted based on the length of the band with the use of a prediction equation.

TABLE 17.2 Examples of Resistance Band Length–Tension Relationships

Length–tension relationships							
Width (mm)	Color	110 CM (43.3 IN.)	120 CM (47.2 IN.)	130 CM (51.2 IN.)	140 CM (55.1 IN.)	150 CM (59.1 IN.)	Tension prediction equation
14	Yellow	2.6 kg (5.7 lb)	5.7 kg (12.6 lb)	8.1 kg (17.9 lb)	9.8 kg (21.6 lb)	11.5 kg (25.4 lb)	$Y = -0.003x^2 + 0.98x - 69.82$
22	Red	4.6 kg (10.1 lb)	9.6 kg (21.2 lb)	13.3 kg (29.3 lb)	16.6 kg (36.6 lb)	19.2 kg (42.3 lb)	$Y = -0.004x^2 + 1.38x - 99.49$

Adapted by permission from D.T. McMaster, J. Cronin, and M. McGuigan, "Quantification of Rubber and Chain-Based Resistance Modes," *Journal of Strength and Conditioning Research* 24, no. 8 (2010): 2061.

32	Blue	8.5 kg (18.7 lb)	14.8 kg (32.6 lb)	19.5 kg (43.0 lb)	23.9 kg (52.7 lb)	27.3 kg (60.2 lb)	$Y = -0.004x^2 + 1.60x - 114.86$
48	Green	11.8 kg (26.0 lb)	16.5 kg (36.4 lb)	24.0 kg (52.9 lb)	30.0 kg (66.1 lb)	49.3 kg (108.7 lb)	$Y = -0.007x^2 + 2.43x - 179.56$
67	Black	15.4 kg (34.0 lb)	29.1 kg (64.2 lb)	40.0 kg (88.2 lb)	49.3 kg (108.7 lb)	57.2 kg (126 lb)	$Y = -0.010x^2 + 3.73x - 269.21$

Adapted by permission from D.T. McMaster, J. Cronin, and M. McGuigan, "Quantification of Rubber and Chain-Based Resistance Modes," *Journal of Strength and Conditioning Research* 24, no. 8 (2010): 2061.

Similar to how resistance is applied with the use of chains, the use of bands requires the strength and conditioning professional to determine how much load is provided by the free weight and how much is provided from the bands throughout the range of movement (8). Specifically, the strength and conditioning professional must determine the load of the band at the bottom and top portions of the movement and create an average of these two loads. Based on the recommendations of Baker (8), an athlete who wants to train at a 5RM load of 150 kilograms (331 lb) with bands would subtract the average of the two band positions (i.e., top and bottom) and reduce the load on the bar by that amount. So if the band exerts a zero resistance load at the bottom of the movement and a load of 26.6 kilograms (58.6 lb) at the top position (i.e., lockout in the bench press), based on the length of the band achieved as it is stretched, the athlete would subtract 13.3 kilograms (29.3 lb) from the total weight on the bar without the bands to get 136 to 137 kilograms (300-302 lb).

Applying Resistance Bands to Free Weight Exercises

To apply resistance bands to free weight resistance exercises, one can use a multitude of methods. The band may be attached to the barbell and either a customized attachment point on a squat rack or heavily weighted dumbbells (8). When the bands are applied, the

highest tension and the total resistance load are provided to the athlete at the top position. Conversely, at the bottom position the applied load is reduced, because the bands will no longer be stretched and thus no longer apply resistance to the barbell. For example, in the squat at the lowest point (i.e., bottom position, [figure 17.3](#)), the bands are slack and do not actively create tension on the barbell, thus adding no additional resistance to the exercise. As the athlete ascends from the bottom position and approaches the lockout position, the bands will impart a larger stretch resistance, and the athlete will experience the full load of both the bar and the bands ([figure 17.4](#)). It is important to note that as the athlete ascends from the bottom position the stretch load gradually increases. Conversely, as the athlete descends, the stretch load gradually decreases.



FIGURE 17.3 Bottom position of a squat performed with bands.



FIGURE 17.4 Top position of a squat performed with bands.

Application of an Accentuated Eccentric Load

The prescription of training loads during resistance training typically employs the use of a constant load that is lifted during the eccentric and concentric portions of the resistance training exercise. This practice creates a unique loading problem because it is well documented that muscles are stronger during eccentric (i.e., lengthening) versus concentric (i.e., shortening) muscle actions, resulting in what is referred to as an *eccentric strength deficit* ([figure 17.5](#)) (51, 91). In fact, depending on the exercise, up to 50% more force can be produced during eccentric muscle actions when compared to concentric muscle actions (34, 53, 64, 70, 143). As

such, the use of constant loads during resistance training results in a lower relative load being used during the eccentric phase of the movement when compared to the concentric phase (144). As a result of this submaximal loading, motor unit recruitment and firing rates during the eccentric phase of the movement may be reduced (125), sarcoplasmic calcium release may decrease, and stimulus for myocellular adaptations may be reduced (42). To address these issues, alternative training methods applying load during resistance training are often sought so that loads are applied in a phase-specific manner when targeting hypertrophic adaptations and increases in maximal strength (144).

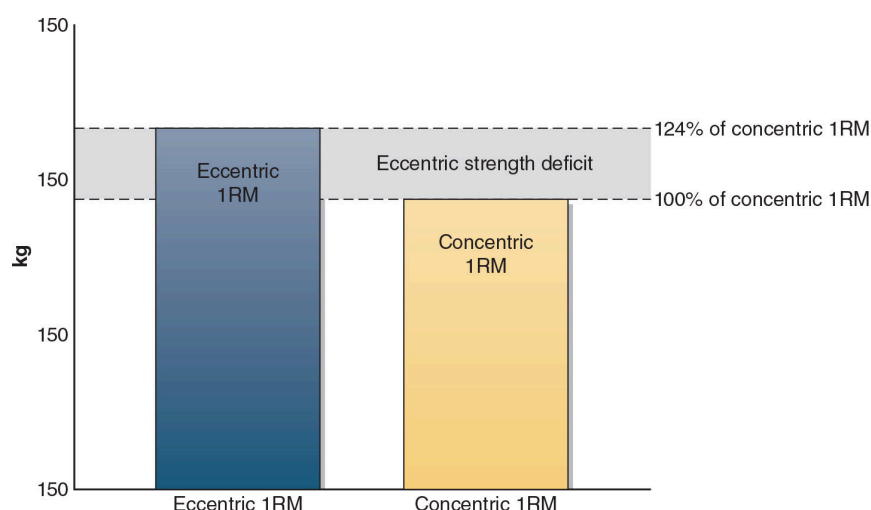


FIGURE 17.5 Example strength deficit between the eccentric and concentric 1-repetition maximum (1RM) in the bench press.

Data from Kelly et al. (2015).

One variable resistance training strategy that addresses this differential is the use of **accentuated eccentric loading (AEL)**, which is sometimes referred to as **augmented eccentric loading** (95, 132). With this loading strategy, an additional load is applied during the eccentric phase of the movement and then removed during the concentric phase of the movement. For example, an eccentric muscle action can be overloaded during a countermovement jump by holding dumbbells and releasing them

before the initiation of the vertical jump (122). Alternatively, if the athlete's 1RM bench press is 120 kilograms (265 lb), the eccentric phase of the movement could be performed with 100% of the concentric 1RM (i.e., 120 kg [265 lb]) and the load can be reduced to 60% of 1RM (i.e., 72 kg [159 lb]) during the concentric phase of the exercise (32). Typically, this is accomplished with the use of weight releasing devices, often referred to as a *weight releaser* (91, 95).

Benefits Associated With Accentuated Eccentric Loading

The primary benefits associated with the use of AEL occur as a result of an overload applied during the eccentric phase of a resistance training or jumping exercise (91). Relatively little research has explored the chronic application of AEL, but due to the greater mechanical tension (43) and work (142, 143) during the eccentric phase, some researchers have suggested that AEL would be useful when targeting hypertrophy. While this rationale is sound, based on the available research increases in muscle size associated with AEL are comparable to traditional training methods (78, 119). It is, however, plausible that AEL results in architectural changes, such as the elongation of the muscle fascicles, allowing for higher shortening velocities which could affect strength and power performance (91). Additionally, the application of AEL tactics may have further benefits to enhancing strength and power performance by inducing changes to the neuromuscular system (144). This hypothesis is partially supported by the work of Walker and colleagues (144), who have reported improved voluntary muscle activation and significantly greater improvements in maximal strength after 10 weeks of training with AEL when compared to traditional resistance training. Ultimately, while there is an emerging body of evidence that using AEL tactics stimulates greater improvements in maximal strength (52, 144) and jump performance (141), more research is needed to determine the most effective AEL programming strategies.

Programming and Accentuated Eccentric Loading

When implementing AEL as part of a periodized resistance training program it is important that the strength and conditioning professional remember that AEL is an advanced training tactic that is used to exploit the established benefits of eccentric overload training (131). As such, this training tactic should be used with caution, and it is important that the athlete's resistance training experience and strength levels are considered when determining the exercises that AEL is implemented with and the loads used during both the eccentric and concentric phases of the movement (91).

Accentuated eccentric loading (AEL) is an advanced training tactic and should be reserved for athletes with extensive resistance training experience.

When programming AEL exercises the magnitude of the eccentric load is generally considered as either being submaximal (i.e., <100% of concentric 1RM), maximal (i.e., 100% of concentric 1RM), or supramaximal (i.e., >100% of concentric 1RM) (91, 143). The decision about which eccentric load to apply is likely mitigated by the targeted performance outcome and by the athlete's training experience and their relative strength levels, with stronger athletes likely experiencing greater benefits from training with AEL tactics (91).

One approach for implementing submaximal AEL is to apply this loading tactic with kettlebells, dumbbells, or elastic bands while performing plyometric exercises (e.g., countermovement, box, drop, and depth jumps) ([figure 17.6](#)) (1, 91). For example, when a 20-kilogram (44 lb; ~20% of body mass) load is applied with dumbbells that are held during the eccentric phase and released prior to the concentric phase of a countermovement jump, jump performance is acutely enhanced (122). Similarly, when an eccentric load that is 20% to 30% of body mass is applied during the eccentric phase of a drop jump and released before the concentric phase of the jump,

performance is improved (1). Ultimately, based on the available evidence, loads of 20% to 30% of body mass should be implemented during AEL when attempting to potentiate performance during the concentric phase of a plyometric jumping activity (91).

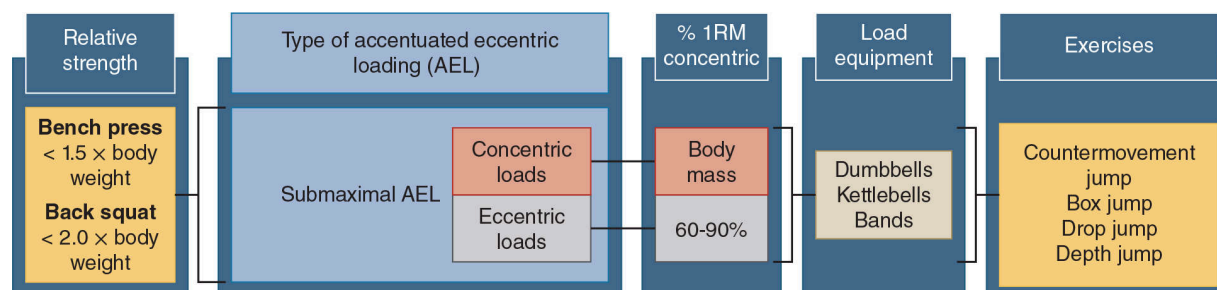


FIGURE 17.6 Accentuated eccentric loading prescription recommendations for plyometric exercises.

Adapted by permission from J.J. Merrigan, J. Borth, C.B. Taber, et al., "Application of Accentuated Eccentric Loading to Elicit Acute and Chronic Velocity and Power Improvements: A Narrative Review," *International Journal of Strength and Conditioning* 2, no 1. (2022): 12. Distributed under the terms of the Creative Commons Attribution 4.0 International License (<http://creativecommons.org/licenses/by/4.0/>).

The application of maximal and supramaximal AEL are best suited for multijoint resistance training exercises, such as the bench press, back squat, and front squat with the use of either weight releasers or computer-driven devices ([figure 17.7](#)) (92, 98, 136). For example, Taber and colleagues (136) report significant improvements in concentric velocity with loads between 30% and 80% of the 1RM during the bench press when a maximal AEL (100% of the 1RM) is implemented. Conversely, when supramaximal AEL is applied with loads between 105% and 120% of the concentric 1RM, slower concentric movement velocities occur during the repetition when the concentric phase is performed with loads between 30% and 80% of the 1RM (25, 78, 91-93, 142). However, it is plausible that the differential between the AEL and the load lifted during the concentric phase of the movement mitigates the acute performance gains. For example, if a differential >40% between the AEL and the concentric load is programmed, significant acute increases in velocity and power occur during the concentric phase of the repetition (92).

Additionally, when this type of loading is applied longitudinally, significantly greater increases in maximal strength occur (144).

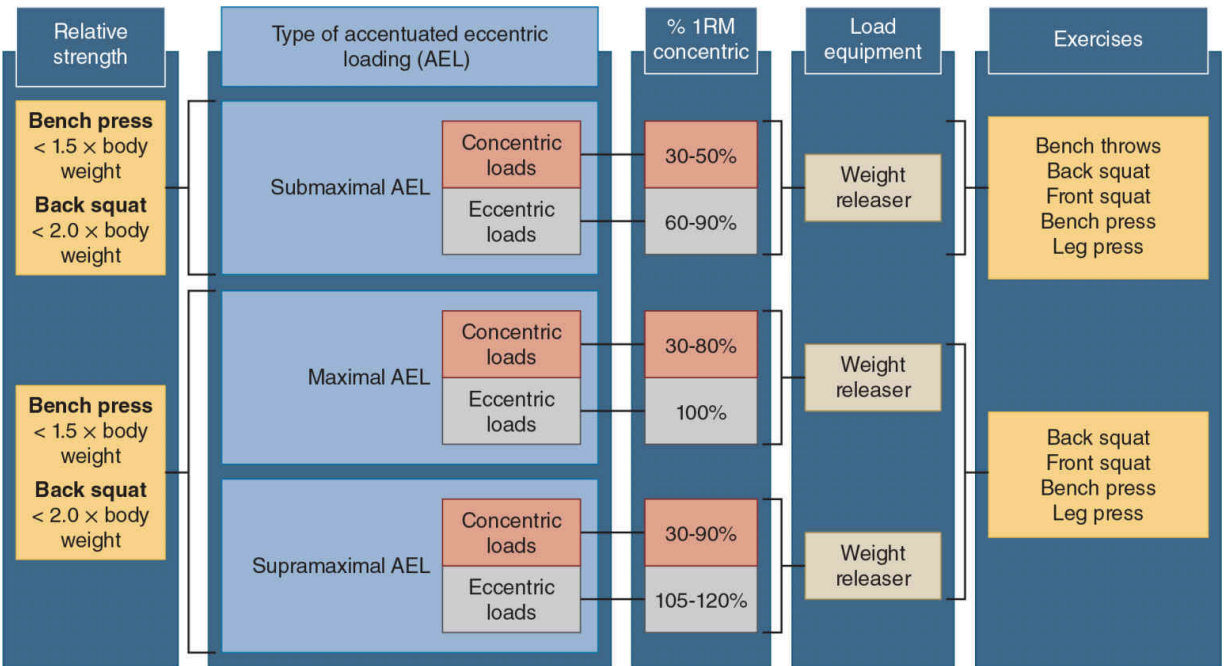


FIGURE 17.7 Accelerated eccentric loading (AEL) prescription recommendations for resistance training exercises.

Adapted by permission from J.J. Merrigan, J. Borth, C.B. Taber, et al., "Application of Accelerated Eccentric Loading to Elicit Acute and Chronic Velocity and Power Improvements: A Narrative Review," *International Journal of Strength and Conditioning* 2, no 1. (2022): 12. Distributed under the terms of the Creative Commons Attribution 4.0 International License (<http://creativecommons.org/licenses/by/4.0/>).

When programming AEL, three basic strategies can be implemented: using AEL on every repetition contained within the set, only applying AEL on the first repetition of the set, or using AEL on the first repetition of a cluster set ([figure 17.8](#)) (51). When AEL is applied on every repetition of a set the set must be structured with a cluster set format (i.e., 15-40 s interrepetition rest) so that the weight releaser can be reapplied after every repetition ([figure 17.8a](#)). The rest interval applied should be structured to account for the time needed to reapply the weight releaser and to re-rack and un-rack the bar. Alternatively, the AEL can be applied on the first repetition of the set, which allows the repetitions contained in the set to be performed in a continuous manner with enhanced

movement velocity occurring during the later repetitions contained within the set ([figure 17.8b](#)) (51). For example, Merrigan and colleagues (92) report that when supramaximal loads are applied on only the first repetition of a set of 5 the velocity of movement during repetitions 3 and 5 is significantly enhanced. Based on these data, a hybrid strategy could be employed where a series of cluster sets of 3 repetitions could be performed with the AEL applied on the first repetition and 25 to 35 seconds of intraset rest applied between clusters ([figure 17.8c](#)). With loading pattern, a greater density of AEL can be applied while still getting acute performance enhancement during the third repetition of each cluster (51, 92, 101).